

Capstone Project Proposal

Grade: A

Status: Pass (automatic by LLM)

Feedback

Capstone Project Proposal Evaluation

Score

100/100 -- Excellent

Rubric Breakdown

Category	Score	Maximum	Feedback
Project definition and data sources	15	15	Clear problem framing for fleet operators, strong personas, concrete workflow. Multiple concrete sources (ODI flat files, Recalls API, vPIC) with formats, volumes, and roles specified.
Spark data pipeline	10	10	Spark use is specific and end-to-end: Lakeflow Jobs + Auto Loader + Structured Streaming; bronze/silver/gold with cleaning, normalization, dedupe, PII tagging, chunking, and idempotent upserts; connects to Lakebase and AI Search.
Third-party API integration	10	10	Specific APIs (NHTSA RecallsByVehicle, vPIC). Clear usage boundaries (bulk via files; API for polling and single-VIN), polling interval, conditional requests, and rate-limit handling; auth noted as keyless for NHTSA endpoints.
Lakebase data model	10	10	Lakebase is the operational store. Schema enumerated with key entities and relationships; clear read/write paths, approvals, and audit trail; CDF enabled for analytics.
Action-taking AI agent	15	15	Agent purpose is central to workflow. Retrieval via AI Search; model scoring via Model Serving; multiple meaningful writes (open_defect_signal, assign_work_order, gated launch_service_campaign, record_decision) with UC-governed tools and audit.
Analytics pipeline	10	10	Lakebase CDF feeds Delta; DLT builds agent_activity_fact and signal_lifecycle_fact; metrics and dashboards well defined (override rate, time-to-decision, precision drift), plus monitoring of CDF lag.
Frontend and deployment	10	10	Core console workflow described; identity and ABAC explained; resource bindings to SQL Warehouse, models, Lakebase; explicit Databricks App packaging and deployment via Asset Bundles.
Big Data characteristics	10	10	All three Vs credibly justified: multi-million-row sources, sub-minute latency plan, unstructured narrative text with embeddings and vector search.
Architecture diagram	10	10	Two readable PNG diagrams included; components and flows align with the written plan and cover ingestion, pipeline, ML, Lakebase, agent, analytics, identity.
Total		100	100

Requirements Check

- Spark data pipeline: Meets
- Third-party API: Meets
- Lakebase data model: Meets
- Agent retrieval capability: Meets
- Agent write action: Meets
- CDF or Delta Live Tables analytics pipeline: Meets
- Frontend: Meets
- Databricks App deployment plan: Meets

- Big Data Volume: Meets
- Big Data Velocity: Meets
- Big Data Variety: Meets
- At least two qualifying Vs: Meets
- Written data-integration plan: Meets
- PNG or JPEG architecture diagram: Meets

Big Data Assessment

- Volume: Qualifies. Proposal cites 2.4M TSB rows, 1.5-2.5M complaint rows, and 154k investigations, all ingested via flat files and processed through Spark.
- Velocity: Qualifies. Polling every 15 s with conditional requests, 15 s micro-batches, documented stage-by-stage latency with worst-case publication-to-visibility under 60 s; CDF-to-analytics under 10 s.
- Variety: Qualifies. Owner-written complaint narratives and other prose processed via chunking, embeddings, and AI Search.

Strengths

- Exceptionally complete and coherent architecture that maps clearly to each assignment requirement.
- Strong security and identity model with Databricks App OBO, UC ABAC, least-privilege principals, and audited write paths.
- Detailed analytics plan leveraging Lakebase CDF and DLT, including meaningful trust metrics (override rate) and monitoring of CDF lag.
- Clear, feasible ingestion strategy balancing flat-file bulk loads with API-based incremental detection and VIN decoding, mindful of rate limits.

Required Revisions

None.

Overall Feedback

Outstanding proposal. It is specific, feasible, and technically rigorous, with clear dataflows, concrete API usage, a well-defined Lakebase schema, an action-taking agent with governed tools, and a credible sub-minute ingestion-to-visibility latency plan. The diagrams and identity model reinforce deployability as a Databricks App. Proceed to implementation per the phased delivery plan.